# CS & IT ENGINEERING




## Algorithms

### Analysis of Algorithms




Lecture No.- 10

By- Aditya sir

# Recap of Previous Lecture



- Topic
- Topic

Questions

TC analysis of Recursive Algo.

# Topics to be Covered



- Topic
- Topic
- Topic

TC analysis of Recursive Algo
PYQs + Practice Questions

Tomorrow Sat → no lec

Lec → Sunday afternoon. (tentative 1PM)



# About Aditya Jain sir

1. Appeared for GATE during BTech and secured AIR 60 in GATE in very first attempt - City topper
2. Represented college as the first Google DSC Ambassador.
3. The only student from the batch to secure an internship at Amazon. (9+ CGPA)
4. Had offer from IIT Bombay and IISc Bangalore to join the Masters program
5. Joined IIT Bombay for my 2 year Masters program, specialization in Data Science
6. Published multiple research papers in well known conferences along with the team
7. Received the prestigious excellence in Research award from IIT Bombay for my Masters thesis
8. Completed my Masters with an overall GPA of 9.36/10
9. Joined Dream11 as a Data Scientist
10. Have mentored 12,000+ students & working professions in field of Data Science and Analytics
11. Have been mentoring & teaching GATE aspirants to secure a great rank in limited time
12. Have got around 27.5K followers on Linkedin where I share my insights and guide students and professionals.

Telegram




Telegram Link for Aditya Jain sir: https://t.me/AdityaSir_PW



# Topic : Analysis of Algorithms

1.
```
Algo AJ(n)
{
    if (n ==1)
        return 1
    else
        return (AJ (n-1) + AJ(n-1))
}
```

T(n)

b

T(n-1) + T(n-1)

$n \longrightarrow 2^n$

AJ(n-1) → O(n)

AJ(n-1) + AJ(n-1) → O(2^n)



# Topic : Analysis of Algorithms

Step-1 Recurrence Relation

$T(n) = b, n = 1$ → B.C

$T(n) = T(n-1) + T(n-1) + a, n \geq 1$

$= 2T(n-1) + a, n \geq 1$



# Topic : Analysis of Algorithms

Step-2 Solve Recurrence

$T(n) = 2T(n-1) + a \rightarrow .....(1)$

$T(n-1) = 2T(n-2) + a$

$T(n) = 2[2T(n-2) + a] + a$

$= 2^2T(n-2) + 2a + a$

$= 2^2T(n-2) + 3a$ —— (2)

$T(n-2) = 2T(n-3) + a$

$3 = (2^2 - 1)$

$T(n) = 2^2(2T(n-3) + a) + 3a$

$= 2^3T(n-3) + 2^2a + 3a$

$= 2^3T(n-3) + 4a + 3a$

$= 2^3T(n-3) + 7a$

$= 2^3T(n-3) + (8-1)\,a$ —— (3)



# Topic : Analysis of Algorithms

## Generalized form

$$T(n) = 2^k (n-1) + (2^k - 1)\, a$$

For B.C., $(n-k) = 1$

$$k = (n-1)$$

$$T(n) = 2^{n-1} T(1) + (2^{(n-1)} - 1)a$$

$$= \frac{2^n}{2} \times 1 + \left(\frac{2^n}{2} - 1\right) a$$

Dominating term.

$$2^{n-1} = 2^n \times 2^{-1}$$

$$= 2^n \times \frac{1}{2}$$

$$= \frac{2^n}{2}$$



# Topic : Analysis of Algorithms

Step-3: Apply Asymptotic Notation

$T(n) = O(2^n)$



# Topic : Analysis of Algorithms

2.

```
Algo AJ(n)
{
    if (n ==2)
        return 2
    else
        return AJ (√n)
}
```

T(n)

b

T(√n)

$\longrightarrow O(\log(\log n))$



# Topic : Analysis of Algorithms

Step.1 Find the recurrence

$T(n) = b, \quad n = 2$

$T(n) = T(\sqrt{n}) + a, \quad n > 2$

Base Condition



# Topic : Analysis of Algorithms

Step.2 Solve recurrence using back substitution

$T(n) = T(\sqrt{n}) + a$

$T(n) = T(n^{1/2}) + a$.....(i)

$\sqrt{n} = n^{1/2}$

$T(n^{1/2}) = T(n^{1/4}) + a$

$T(n) = (T(n^{1/4}) + a) + a$

$= T(n^{1/4}) + 2a$

~~$= T(n^{1/2}) + 2a$~~

$T(n) = T(n^{1/2^2}) + 2a$ — (2)

$\sqrt{\sqrt{n}}$

$(n^{1/2})^{1/2} = n^{1/2^2}$

$T(n) = T(n^{1/4}) + 2a$

$T(n^{1/2}) = T(n^{1/2^3}) + a$

$T(n) = (T(n^{1/8}) + 3a$

$= T(n^{1/2^3}) + 3a$.....(2)

~~$= T(n^{1/2}) + 2a$~~



# Topic : Analysis of Algorithms

Generalized from

$T(n) = T(n^{1/2^k}) + k * a$

For base condition

$n^{1/2k} = 2$

$\log_2\left(n^{\frac{1}{2k}}\right) = \log_2^2$

$\frac{1}{2k}\log_2^n = 1$

$2^k = \log_2^n$

$k = \log_2(\log_2^n)$

$T(n) = T(2) + k * a$

$T(n) = b + \log_2(\log_2 n) * a$

Value of Rec

Step.3- Apply Asymptotic notation

$T(n) = O(\log(\log n))$

$$n^{1/2^k} = 2$$

Take $\log_2(\,)$

$$\log_2(n^{1/2^k}) = \log_2 2$$

$$\frac{1}{2^k}\log_2 n = 1$$

$$\log_2 n = 2^k$$

$$\log_2(\log_2 n) = \log_2(2^k) \Rightarrow k = \log(\log n)$$



# Topic : Analysis of Algorithms

#Q.

```
Algo AJ(n)
{
    if (n==2)
        return 2
    return (AJ(√n) + AJ (√n))
}
```

$T(n)$

b

Time taken $T(\sqrt{n}) + T(\sqrt{n})$

(most correct → tight UB) ✓

A $O(n)$

B $O(\sqrt{n})$

C $O(\log n)$

D $O(\log(\log n))$

Step 1 :

$$T(n) = \begin{cases} T(n) = b, & n = 2 \\ T(n) = a + T(\sqrt{n}) + T(\sqrt{n}), & n > 2 \\ T(n) = 2T(\sqrt{n}) + a, & n > 2 \end{cases}$$

$T(n) = 2T(\sqrt{n}) + a$

$T(n) = 2T(n^{1/2}) + a$ — ①

$T(n^{1/2}) = 2T(n^{1/2^2}) + a$

$T(n) = 2\left[2T(n^{1/2^2}) + a\right] + a$

$= 2^2 T(n^{1/2^2}) + 2a + a$

$T(n) = 2^2 T(n^{1/2^2}) + 3a$ — ②

$T(n^{1/2^2}) = 2T(n^{1/2^3}) + a$

$3a = (2^2 - 1)a$

$T(n) = 2^2\left[2T(n^{1/2^3}) + a\right] + 3a$

$= 2^3 T(n^{1/2^3}) + 2^2 a + 3a$

$T(n) = 2^3 T(n^{1/2^3}) + 7a$ — ③

$(8-1)a = (2^3 - 1)a$

general term

$T(n) = 2^k * T(n^{1/2^k}) + (2^k - 1) * a$

for base Condition, $n^{1/2^k} = 2$

$\log(n^{1/2^k}) = 1$

$\log n = 2^k$

$T(n) = 2^k T(2) + (2^k - 1) * a$

$T(n) = \log n * b + (\log n - 1) * a$

Step3 :- $AJ(n) \rightarrow O(\log_2 n)$



# Topic : Analysis of Algorithms

#Q. Algo ~~AJ(n)~~ AJ1(n) → T(n)

Assum AJ2(n) → O(1)

```
{
    if (n==1)
        return 2
    else
    return (AJ 1(n/2) + AJ1(n/2) + AJ2(n))
    }
}
```

AJ1(1) → 2

b

T(n/2) T(n/2) c

A $O(\sqrt{n})$

B $O(n)$

C $O(n^2)$

D $O(\log n)$

Step 1 :- $T(n) = b, \quad n = 1$

$$T(n) = a + T(n/2) + T(n/2) + c, \quad n > 1$$

$$= 2T(n/2) + (a + c)$$

$$T(n) = 2T(n/2) + d, \quad n > 1$$

$T(n) = 2T(n/2) + d$ — (1)

$T(n/2) = 2T(n/2^2) + d$

$T(n) = 2^2 T(n/2^2) + 3d$

$T(n/2^2) = 2T(n/2^3) + d$

$T(n) = 2^3 T(n/2^3) + 2^2 d + 3d$

$T(n) = 2^3 T(n/2^3) + 7d$

General,

$T(n) = 2^k T(n/2^k) + (2^k - 1)d$

For B.C, $n/2^k = 1$

$n = 2^k$

$$T(n) = 2^k T(n/2^k) + (2^k - 1) * d$$

$2^k = n$

$$T(n) = n * T(1) + (n-1) * d$$

$$T(n) = n * b + (n-1) \times d$$ ⟶ value of Recurrence

Step3 :- TC of AJI(n) = O(n) ✓



# Topic : Analysis of Algorithms

#Q.

```
Algo AJ(n)
{
    if (n ==1)
        return 1
    else
    {
        AJ (n/2)
        AJ (n/2)
        AJ 2 (n)
    }
}
```

V.V. Imp

b

T(n)

T(n/2)

T(n/2)

O(n)

40%

A. O(n)

B. O(n logn)

C. O(logn)

D. $O(n^2 \log n)$

$T(n) = b, \quad n = 1$

$T(n) = a + T(n/2) + T(n/2) + O(n)$

$= 2T(n/2) + O(n) + a$

$O(n) \rightarrow n$

$T(n) = 2T(n/2) + n, \quad n > 2$

$T(n) = 2T(n/2) + n$ —— (1)

$T(n/2) = 2T(n/2^2) + n/2$

$T(n) = 2^2 T(n/2^2) + 2 \times n/2 + n$

$T(n) = 2^2 T(n/2^2) + 2n$ —— (2)

$T(n/2^2) = 2T(n/2^3) + n/2^2$

$T(n) = 2^3 T(n/2^3) + n/2^2 \times 2^2 + 2n$

$T(n) = 2^3 T(n/2^3) + 3n$ —— (3)

general,

$$T(n) = 2^k \, T(n/2^k) + k * n$$

for Base Condition.

$$n/2^k = 1 \longrightarrow 2^k = n \longrightarrow (k = \log_2 n)$$

$$T(n) = n * T(1) + n * \log_2 n$$

$$\boxed{T(n) = n * b + n * \log_2 n}$$

$$T(n) \Rightarrow \boxed{O(n \log_2 n)}$$



# Topic : Analysis of Algorithms

Tricky

#Q. Given a recurrance, determine the time compleixty.

$T(n) = 2 , n = 2$

$T(n) = \sqrt{n} * T(\sqrt{n}) + n, n > 2$

V. V. Advance

A O(nlogn)

B O(nlog (logn))

C $O(n^2)$

D O(log (logn))

$T(n) = \sqrt{n}\,T(\sqrt{n}) + n$

$T(n) = n^{1/2}\,T(n^{1/2}) + n$ — ①

$T(n^{1/2}) = n^{1/2^2}\,T(n^{1/2^2}) + n^{1/2}$

$T(n) = n^{1/2}\left[n^{1/2^2}\,T(n^{1/2^2}) + n^{1/2}\right] + n$

$= n^{(1/2 + 1/2^2)}\,T(n^{1/2^2}) + n + n$

$T(n) = n^{(1/2 + 1/2^2)}\,T(n^{1/2^2}) + 2n$ — ②

$T(n^{1/2^2}) = n^{1/2^3}\,T(n^{1/2^3}) + n^{1/2^2}$

$T(n) = n^{(1/2^2 + 1/2)}\left[n^{1/2^3}\,T(n^{1/2^3}) + n^{1/2^2}\right] + 2n$

$$\left(\frac{1}{2^2} + \frac{1}{2^2} + \frac{1}{2}\right)$$

$$= \frac{1}{2} + \frac{1}{2} = 1$$

$$T(n) = n^{\left(\frac{1}{2} + \frac{1}{2^2} + \frac{1}{2^3}\right)} T\left(n^{\frac{1}{2^3}}\right) + n^{\frac{1}{2^2}} * n^{\left(\frac{1}{2^2} + \frac{1}{2}\right)} + 2n$$

$$= n^{\left(\frac{1}{2} + \frac{1}{2^2} + \frac{1}{2^3}\right)} * T\left(n^{\frac{1}{2^3}}\right) + n^1 + 2n$$

$$T(n) = n^{\left(\frac{1}{2} + \frac{1}{2^2} + \frac{1}{2^3}\right)} * T\left(n^{\frac{1}{2^3}}\right) + 3n \quad \text{——} \quad (3)$$

general Term

$$T(n) = n^{\left(\underbrace{1/2 + 1/2^2 \cdots + 1/2^k}\right)} * T\left(n^{1/2^k}\right) + k * n \quad \text{——} (4)$$

$$\frac{1}{2} + \frac{1}{2^2} + \frac{1}{2^3} \cdots + \frac{1}{2^k} \longrightarrow \underline{\underline{GP}}$$

$a = 1/2$

$r = 1/2$

$n = k$

$$S_n = \frac{1/2\left(1 - 1/2^k\right)}{1 - 1/2} = \underline{\underline{\left(1 - 1/2^k\right)}}$$

General Term

$$T(n) = n^{(1-1/2^k)} * T(n^{1/2^k}) + k*n \quad -(5)$$

For B.C, $n^{1/2^k} = 2$

$$2^k = \log n$$

$$k = \log(\log n)$$

$$T(n) = \frac{n^1}{n^{1/2^k}} * T(2) + n*\log(\log n)$$

$$= n/2 \times 2 + n*\log(\log n) = \boxed{n + n*\log(\log n)}$$

$$T(n) \rightarrow O(n \log \log n)$$



# Topic : Analysis of Algorithms

```
Algo AJ( n)
{
        if (n ==1)
              return 1
        else
              return [AJ(√n) + 10]
}
```

HW1



# Topic : Analysis of Algorithms

```
Algo AJ( n)
{
        if (n ==1)
              return n
        else
              return (AJ (n/2) + 10)
}
```

H.W



# Summary

TC Questions



# THANK - YOU